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PALAEOMAGNETISM AND RADIOMETRIC AGE OF THE JURASSIC CHON AIKE FORMATION FROM SANTA CRUZ PROVINCE, ARGENTINA: IMPLICATIONS FOR THE OPENING OF THE SOUTH ATLANTIC

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At Camarones Bay, the Chon Aike formation of rhyolites, andesites and porphyries exhibits normal remanent magnetization at nine sites and reversed remanence at eight sites A K-Ar age of 166±5 my was obtained from a pair of determinations on each of ten samples This puts the emplacement of these lavas in the Bathonian Stage of the Middle Jurassic The reversed magnetization may therefore correspond to the Pospelovskaja Reversed Zone in the predominantly normal Jurassic geomagnetic field [1] The mean direction of the normal group of sites is the same as that already reported for normally magnetized samples from Puerta Deseado [2], but the reversed group yields a different mean direction A single population formed of the two sets of VGPs falling near the present south poles has a mean at 129°W 82°S (α_{95} = 12°), and when compared with the African mean Jurassic pole could be interpreted in terms of a post Jurassic opening of the South Atlantic

1. Introduction

The only South American Jurassic rocks to have been studied palaeomagnetically until now are the Chon Aike lavas from near Puerta Deseado in the Province of Santa Cruz, Argentina (area PD, inset in fig 1) [2] These rocks are almost exlusively normally magnetized and yield a virtual geomagnetic pole (VGP) at 56°E 84°S The semi-axes of the oval of confidence (P = 0 05) are 7° and 6° When rotated with South America to Africa to form part of Gondwanaland, (pole of rotation 329 4°E 44 0°N, angle of rotation 57° [3]) this VGP plots at 82°E 44°S, a position which is significantly different from that of the African Jurassic pole at 76°E 68°S (α = 18°, N = 5 formations [4]) suggesting that some relative motion between South America and Africa had occurred by the Jurassic [2]

But this conclusion is not consistent with the evidence from sea floor spreading which suggests that the opening up of the South Atlantic Ocean occurred in two stages, the first of which began at about 140 mybp [5] The oldest sediments found off the South Atlantic continental shelves are Upper Jurassic, from near Senegal, while the oldest marine beds in the South Atlantic are Lower Cretaceous Deep sea drilling in the South Atlantic basis indicates a uniform rate of separation of South America and Africa at 2 cm/yr for the last 66 my and if we extrapolate this constant rate we must conclude that separation must have started 130 my ago, i e in the Lower Cretaceous [6] Ladd et al [7] conclude that the initiation of continental drift in the South Atlantic occurred subsequent to about 135 mybp Dietz and Holden [8] associate the actual rifting of West Gondwana (South America plus Africa) with the emplacement of the Serra Geral Basalts in Brazil and of the Kaoka Basalts of South West Africa The K-Ar ages of both these formations range between about 140 and 110 mybp [9, 10]

We now report further palaeomagnetic studies of the Jurassic Chon Aike formation from the vicinity of Camarones (area C in inset to fig 1) in Santa Cruz Province, Argentina.

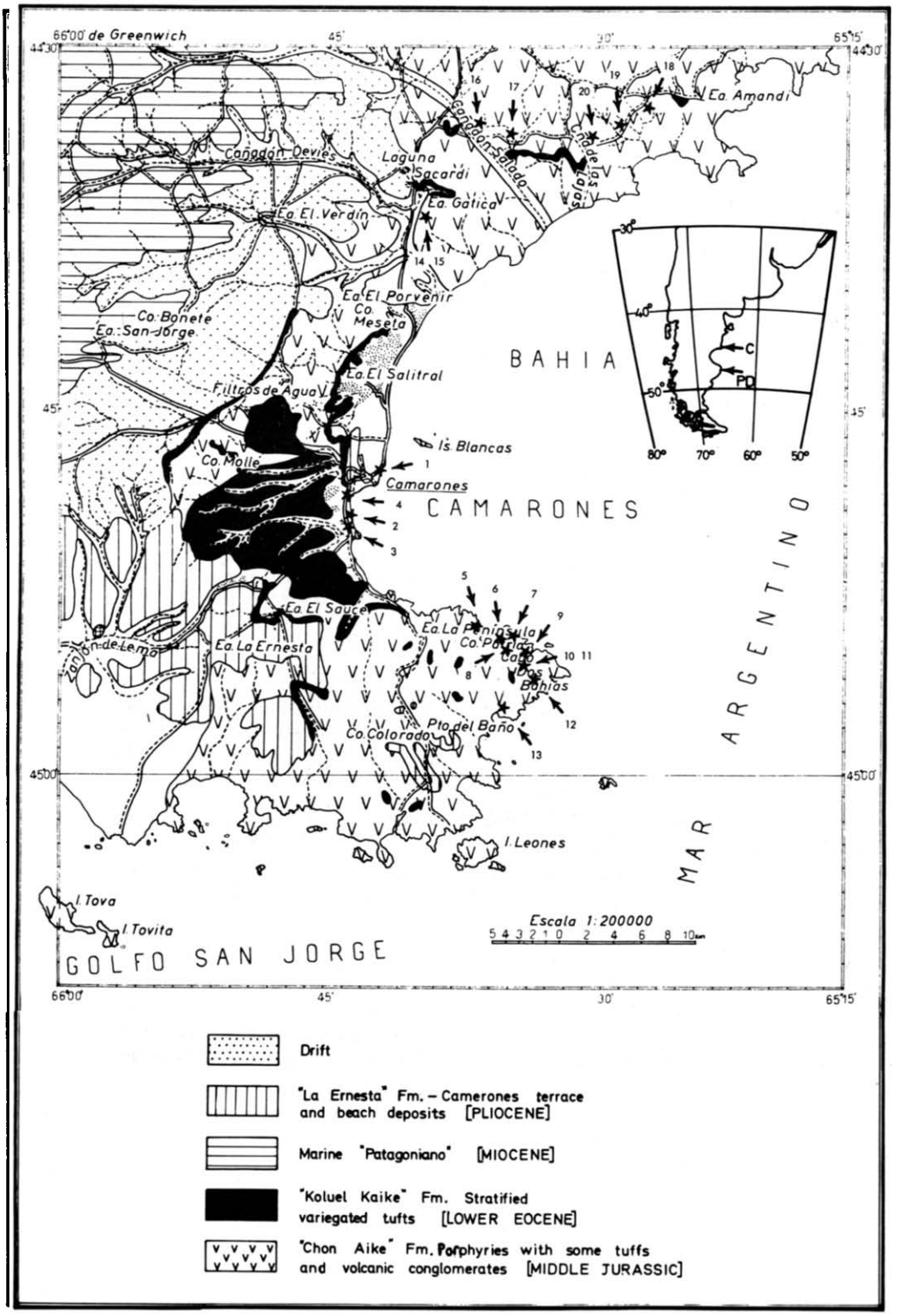


Fig 1 Geological map of the region surrounding Camarones Bay, Santa Cruz Province, Argentina Sampling sites indicated by stars On the inset map, this area is indicated by the point labelled C, and the Puerta Deseado area, sampled by Valencio and Vilas [2] by the point labelled PD

Table 1

Site	Sample numbers	Rock type and description	Samples with stable RM	Polarity of site
1	1–9	Red/grey banded rhyolites	1–9	R
2	10, 11	Banded rhyolites	10, 11	R
3	12, 13	Rhyolitic agglomerate	–	–
4	14, 15	Brick red acid rock beneath bedded volcanic ash	14, 15	R
5	16–19	Orange/brown trachyte	–	–
6	20–23	Yellow/grey quartz-feldspar porphyry weathering to red/brown	21, 22	N
7	24, 25	Similar to site 6	–	–
8	26–28	Similar to site 6	26, 27	R
9	29–35	Grey porphyritic andesite	32, 33	N
10	36–38	Like site 9	36–38	N
11	39–41	Similar to site 10, severely fractured	39, 41	N
12	42–44	Brown porphyry	42, 44	R
13	45–47	Orange/brown porphyry	45, 46	N
14	48–50	Red/brown quartz porphyry with fine grained matrix	49 50	R
15	51–53	Similar to site 14	51, 53	R
16	54–57	Quartz-feldspar porphyry	55	N
17	58–61	Similar to site 16	59, 61	R
18	62–64	Light grey feldspathic porphyry More basic than at other sites	62	N
19	65–67	Similar to site 18	65–67	N
20	68–70	Pinky-grey feldspathic porphyry	68–70	N

2 Geology and geographic location of sampling sites

70 samples from the Chon Aike formation were collected from the vicinity of Camerones Bay in January 1970 by Dr B J J Embleton The sites are marked on the geological sketch map (fig 1) Twenty sites were sampled The rock types are briefly described in table 1 and are mainly rhyolites, trachytes and andesites The sites are distributed to the north and south of Camarones Bay over a distance of about 50 km Most of the samples were highly altered Those suitable for dating were selected according to the criteria outlined by Dalrymple and Lanphere [11] and came mainly from the Cape at the south of Camarones Bay, from sites 6, 8, 9 and 10 The only other sufficiently fresh sample for dating came from site 14, about 15 km north of Camarones Potassium-argon ages were determined using the methods described by Evans et al. [12] The mean isotopic age is 166 ± 5 my (table 2) and the age of 161 my reported by Valencio and Vilas [2] falls into the population of the ages we have measured There is no apparent difference in age between the normally and the reversely magnetized lavas (see tables 1 and 2)

3 Remanent magnetization

Several 1″ cylinders were cut from each hand sample or field drilled core (counted as a sample in table 3) Mean directions were calculated allotting unit weight (a) to site-mean directions and (b) to sample mean directions (table 3) The directions of natural remanent magnetization were highly scattered, but after cleaning by the alternating field method the RM of samples from most of the sites fell into two compact groups, accepted as representing normal and reversed directions of the Jurassic field Site-mean directions of cleaned RM are plotted in fig 2

The procedure adopted for cleaning was as follows

Table 2
Potassium-argon ages from feldspar porphyries, Camarones

K-Ar lab ref no	Sample no	Site no	K_2O (%)	Radiogenic argon content (10^{-2} mm^3 g^{-1})	Atmospheric contamination (%)	Age (my)
231	37	10	6 61 ± 0 03	3 80 ± 0 07	22 5	166 ± 3
				3 78 ± 0 06	21 0	165 ± 3
232	31	9	6 45 ± 0 02	3 72 ± 0 08	48 8	167 ± 4
				3 60 ± 0 06	50 1	162 ± 3
233	38	10	6 33 ± 0 04	3 71 ± 0.05	12 5	169 ± 3
				3 61 ± 0 05	14 6	165 ± 3
247	26	8	6 90 ± 0 03	4 02 ± 0.05	11 8	169 ± 2
				3 80 ± 0 08	61 2	160 ± 4
248	33	9	7 30 ± 0 05	4 14 ± 0 06	27 5	164 ± 3
				4 08 ± 0 06	29.5	162 ± 3
277	21	6	6 08 ± 0 04	3 29 ± 0 05	32 7	157 ± 3
				3 46 ± 0 05	28 0	165 ± 3
278	27	8	7 08 ± 0 04	4 15 ± 0 07	17 3	169 ± 3
				4 18 ± 0 05	14 2	170 ± 3
279	28	8	7 09 ± 0 07	4.22 ± 0 06	15 7	172 ± 3
				4 25 ± 0 06	12 2	173 ± 3
280	35	9	7 21 ± 0 04	4 25 ± 0 06	5 8	170 ± 3
				4 21 ± 0 05	6 2	168 ± 2
281	50	14	6 09 ± 0 05	3.40 ± 0 06	10 0	162 ± 3
				3 29 ± 0 05	8 9	157 ± 3

Mean of all data 166 ± 5 my. $\lambda_B = 4\ 72 \times 10^{-10}$ yr^{-1}, $^{40}K/K = 1\ 19 \times 10^{-2}$ atom%, $\lambda_e = 0\ 584 \times 10^{-10}$ yr^{-1}

Table 3
Palaeomagnetic directions

	N	N_S	D	I	R	CSD	α_{95}
Normal	9		358°	−57°	8 658	17°	11°
magnetization		18	1°	−59°	17 179	18°	8°
Reversed	8		212°	+70°	7 788	14°	10°
magnetization		23	223°	+70°	22 148	16°	6°
Predominantly normal (ref [1])		66	353°	−63°	61 37	–	5°

N = number of sites (counted as unit vectors)
N_S = number of samples (counted as unit vectors)
D, I = declination and inclination of cleaned RM
R = length of resultant vector
CSD = circular standard deviation of distribution of vectors
α_{95} = radius of circle of confidence (at $P = 0\ 05$)
Note The measurements were made on 1 inch cylinders of rock, about three of which were cut from each sample (= field drilled core or hand-sample)

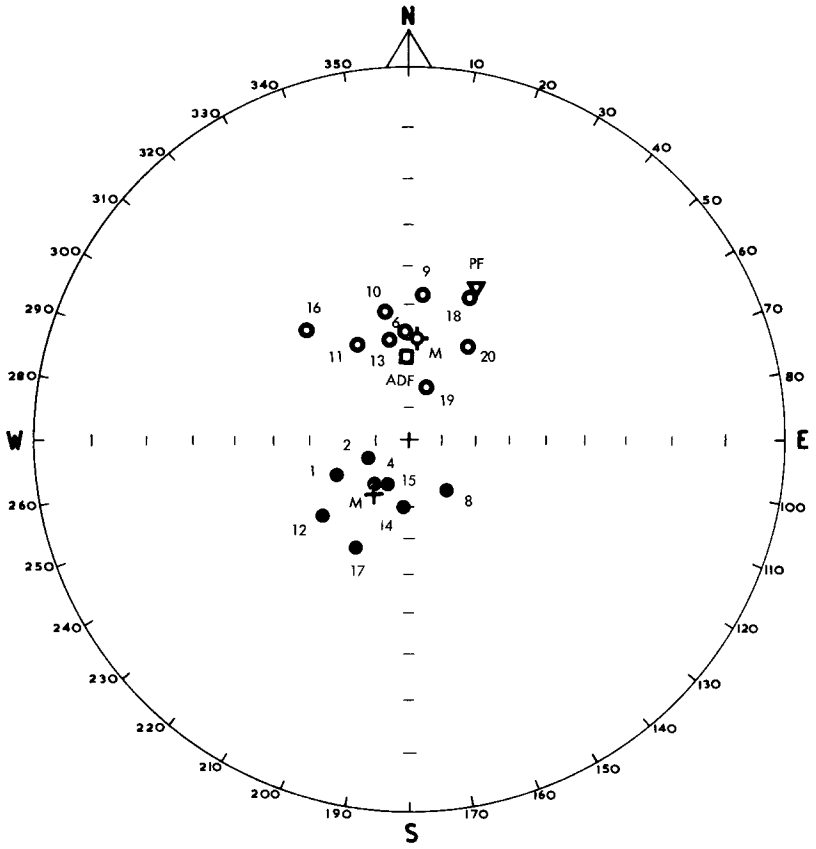


Fig 2 Stereographic projection of site-mean directions of north-seeking RM after cleaning by the alternating field method Unit vectors with upward inclinations represented by open circles those with downward inclinations by black dots The mean direction of the normal group of sites is represented by -◊- and that of the reversed group of sites by + labelled M See table 3 for statistics The axial dipole normal field direction is represented by the open square (ADF) and the present field direction by the open triangle (PF)

AF demagnetization curves were plotted for one or two pilot samples from each site A single peak field value was then chosen for application to the remaining samples from each site The criterion adopted for the choice of this demagnetizing field strength was that the direction of the pilot sample exhibited no further change in response to treatments in higher demagnetizing field strengths apart from anhysteretic effects. In any case, samples with the highest AF coercivities were demagnetized to at least 50% of the natural intensity and less stable samples were demagnetized to about 10% of the natural intensity even if the remanent direction had ceased to change systematically at an earlier stage in the progressive demagnetization of the pilot samples The demagnetizing field strengths used varied from 200 to 500 oersteds depending on the site To test the consistency of the measurements and to confirm that the samples had not been given an anhysteretic remanence while being cleaned, some of the samples were cleaned again in the chosen demagnetizing field and remeasured

4. Other magnetic and petrological observations

The scattered directions of NRM were observed often to be associated with high remanent intensities which were sharply reduced by treatment in alternating fields of about 100 Oe AF coercivities in excess of 400 Oe ($J_I/J_{IO} \approx 0\ 1$) were typical of rock samples

with stable NRM NRM intensities, J_r, varied from about 2000 γ to 10 γ and low field susceptibilities, k, from rather less than 10 γ/Oe to 1500 γ/Oe The Königsberger ratios, Q, varied from 0.002 to about 5 ($Q = J_r/kH$, $H = 0$ 45 Oe)

The hardest remanence was exhibited by the feldspathic porphyry at site 19 for which the AF coercivity was considerably higher than 450 Oe at which $J_r/J_{ro} \approx 0$ 4 and for which the thermomagnetic curve exhibited a Curie point of 670°C The other rocks exhibited some low Curie points (150°–350°C), many rocks exhibited Curie points higher than 500°C, a number exceeding 580°C The rocks showed evidence of low temperature oxidation, the polished sections exhibiting a characteristic red or orange or brown staining throughout The ore grains in samples exhibiting low Curie points were homogeneous Those in samples with high Curie points contained exsolution lamellae of magnetite and ilmenite In some samples, the ore grains were badly corroded frequently oxidation to haematite and rutile was evident

We could not detect any differences in overall petrology or gross magnetic properties of samples with normal and reversed RM Nor were there any systematic differences between the cleaned RM directions of samples with high and low Curie points

5. South pole positions

The mean pole position and statistics of the populations corresponding to the directions listed in table 3 of virtual geomagnetic poles are presented in table 4 We note that the mean pole position obtained from normally magnetized samples and sites is very close to that previously obtained by Valencio and Vilas [2] from their normally magnetized samples from Puerta Deseado However, the site-mean poles obtained from our reversely magnetized samples appear to be displaced from those obtained from normally magnetized samples (fig 3) and the mean of the site-mean normal poles is in fact significantly displaced from the mean of the site-mean reversed poles (table 3)

Three possible explanations of this observation are (1) the RM may have not been completely cleaned and the very slightly 'smeared' distributions of site-mean directions (fig 2) might be indicative of this, (2) the occurrence of unrecognized post-emplacement tectonic movements, and (3) the geomagnetic field 'reversal' may not have been exactly through 180°, neither the main geomagnetic dipole for the period of normal polarity nor that for the period of reversed polarity having been parallel to the geographic axis

As regards (1), it would appear that the small

Table 4
Mean virtual geomagnetic poles falling in southern hemisphere

	N	N_S	Long	Lat	R	CSD	α_{95}
Normally magnetized sites	9		95°E	84°S	8 402	22°	15°
Normally magnetized samples		18	128°E	86°S	16 502	24°	11°
Reversely magnetized sites	8		116°W	68°S	7 501	22°	15°
Reversely magnetized samples		23	121°W	62°S	21 005	25°	10°
All sites	17		129°W	82°S	15 441	26°	12°
All samples		41	127°W	75°S	36 255	28°	8°
Normally magnetized samples (ref [1])		66	56°E	84°S	–	–	6°, 7°

N = number of sites
N_S = number of samples
R = length of resultant vector
CSD = circular standard deviation
α_{95} = radius of circle of confidence ($P = 0$ 05)

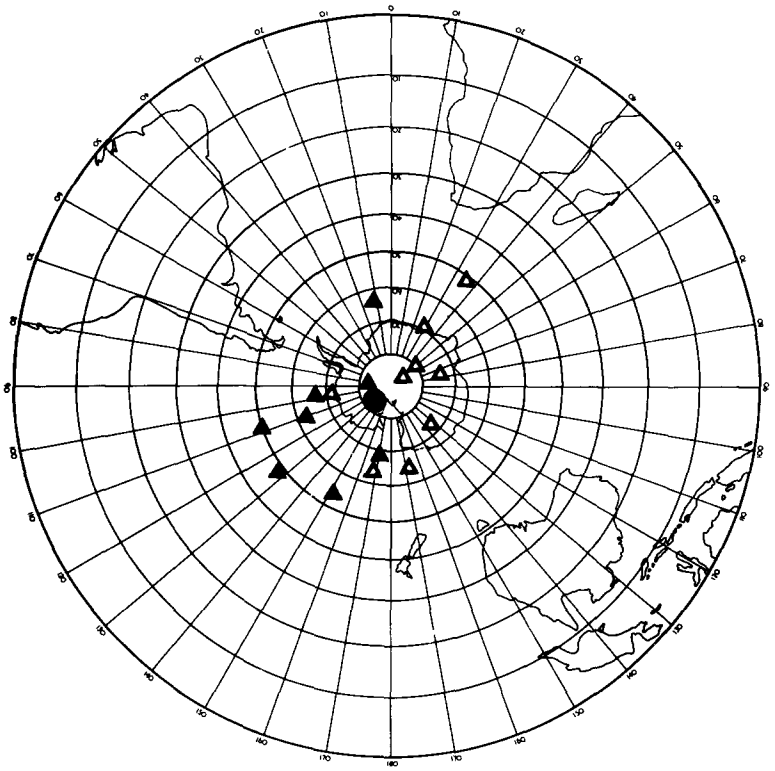


Fig 3 Virtual geomagnetic poles derived from the site-mean RM directions plotted in fig 2 VGPs of normal RM vectors represented by open triangles, those of reversed RM directions by black triangles The mean of all these VGPs is represented by the larger black dot This is accepted as an estimate of the position of the south geographic pole at about 166 mybp

amount of smearing results from the data from site 8 (reversed) and from site 16 (normal) The remaining 15 site-mean directions form two circular populations which are still clearly not anti-parallel (see fig 2), so we may discard this possibility With regard to (2), we did not apply structural corrections because of lack of relevant published data. But in the field it appeared that no significant corrections would be required. Even if future geological mapping reveals that structural corrections are necessary, however, it would be surprising to find that they were such as to rotate all the normal RM directions in one sense and all the reversed RM directions in the opposite sense by the precise amounts required to bring them into exact opposition Nevertheless, this remains a possible explanation. To prove explanation (3), it is necessary to obtain normal and reversed data from other formations in which the same reversal is recorded [1], from a wide geographical area, preferably from localities in other continents.

We consider, however, that the average of the normal and reversed populations of site mean poles (fig 3) must yield a better estimate of the Middle Jurassic (ca 166 mybp) geographic south pole than either one of these populations or the Valencio and Vilas [2] population of normal RM directions.

A considerable number of palaeomagnetic studies have been made on Jurassic rocks from southern Africa. Of these, the Mateke Hills Complex is closest in age (K-Ar age 177±4 my) to the Chon Aike formation. The RM is reversed and the pole is situated at 80°E 59°S, $\alpha_{95} = 8°$, (entry J16 in table 5, ref [4]) Also, the Karoo dolerites have been dated by K-Ar at between 190 and 154 my [13] and have RM of mixed polarity yielding an average pole at 75°E 65°S, $\alpha_{95} = 12°$ (entry J14, table 5, ref [4]). These pole positions do not differ significantly from that obtained from the Chon Aike formation which, when rotated with

South America to Africa so as to reconstitute Gondwanaland [3] is situated at 77°E 58°S, $\alpha_{95} = 12°$

Thus this new pole position can be interpreted in terms of a post Middle Jurassic opening of the South Atlantic

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